

Analysis of the conductivity and current-voltage characteristics nonlinearity in LiNbO₃ crystals of various compositions at temperatures 300–450 K

A.V. Yatsenko^a, S.V. Evdokimov^a, M.N. Palatnikov^{b,*}, N.V. Sidorov^b

^a V.I. Vernadsky Crimean Federal University, Prospekt Vernadskogo 4, Simferopol 295007, Russia

^b Tananaev Institute of Chemistry – Subdivision of the Federal Research Centre “Kola Science Centre of the Russian Academy of Sciences”, 184209, Murmansk Region, Apatity City, Russia

ARTICLE INFO

Keywords:

Lithium niobate
Dopants
Conductivity
Electrical properties
Coercive field
Current-voltage characteristic

ABSTRACT

This paper is dedicated to a study of conductivity and current-voltage characteristics (I–V curve) of lithium niobate (LN, LiNbO₃) single crystals. The studied crystals have been turned to a single domain state. We have researched nominally pure congruent (CLN) and strongly doped LiNbO₃:Mg and LiNbO₃:Zn crystals at $T = 300\text{--}450$ K. Magnitude, activation energy and type of conductivity have been established in CLN, LiNbO₃:Mg and LiNbO₃:Zn in the temperature range $T = 300\text{--}450$ K. Coercive field in LiNbO₃:Mg and LiNbO₃:Zn crystals is much lower than in CLN. However, we have determined that nonlinearity of I–V curve exists in CLN crystals and is absent from LiNbO₃:Mg and LiNbO₃:Zn crystals at temperatures up to 450 K and external electric field strength up to 12 kV/mm. The paper suggests possible explanation for the observed effects.

1. Introduction

Lithium niobate (LN, LiNbO₃) crystals are the classical representative of high Curie temperature displacement ferroelectrics. These crystals both pure and doped are widely used as active elements of opto- and acoustoelectronics [1]. LN crystals with a suppressed photorefractive effect are widely studied nowadays [2]. Near-stoichiometric LN (NSLN) crystals and doped by magnesium (with a content MgO ≥ 4.7 mol% in the melt) and zinc (with a content ZnO ≥ 6.8 mol% in the melt) have a low photorefractive effect [2].

It is known that an increase in Mg or Zn content in crystals leads to a significant decrease in the coercive field E_c . In nominally pure congruent LN (CLN) E_c at room temperature is 24 kV/mm; in LiNbO₃:Mg (LN:Mg) crystals with 5.0 mol% MgO in a crystal is ~ 6 kV/mm [3]; in LiNbO₃:Zn (LN:Zn) crystals with analogous content of ZnO in a crystal is due to different data 3.7–7 kV/mm [4,5]. E_c significantly decreases with an increase in temperature: a LN sample with 5.0 mol% MgO has $E_c \cong 1.5$ kV/mm [6] or $E_c \cong 2.2$ kV/mm [7] at $T = 450$ K. This is why LN:Mg and LN:Zn crystals are preferred for the nonlinearly active laser media with periodically polarized domain structures of micron and submicron sizes [8].

LN crystal is supposed to have a nonlinearity on current-voltage

characteristic (I–V curve) under an external electric field E applied in the direction opposite to the polar axis Z (or the appearance of a pyroelectric nature field). It should happen when E approaches E_c . This is a consequence of a spontaneous polarization switch. Indeed, works [9,10] have demonstrated that I–V curve of CLN shows a significant nonlinearity at as low as $E \geq 0.2$ kV/mm at $T = 523$ K. A qualitatively similar effect was observed in CLN at lower temperatures $T \cong 408$ K [11].

If the nonlinearity of the I–V curves in LN:Mg and LN:Zn crystals is connected with spontaneous polarization switching, than it should manifest itself in LN:Mg and LN:Zn crystals at lower values of E than in CLN crystals. However, I–V curves at high E values are poorly studied in Mg and Zn doped LN. It is only known that I–V curves of LN:Mg (5.0 mol%) are slightly asymmetric with respect to the polarity of the applied voltage at $T = 433$ K [12]. However, there were no signs of nonlinearity at $|E| \leq 0.4$ kV/mm. LN:Mg (5.0 mol%) crystal's I–V curve becomes nonlinear at $E \geq 3$ kV/mm at room temperature after the crystal's treatment with ³He ions with energy 41 MeV [13]. One can evaluate studied sample conductivity at room temperature due to data from [13]. The conductivity turned out to be $1.6 \cdot 10^{-11} (\Omega \cdot \text{cm})^{-1}$. This value is significantly higher than in “as grown” LN crystals. It is characteristic only of LN samples treated by a reduction annealing in a nitrogen-hydrogen gas mixture [14] or a pure hydrogen atmosphere [15]. In

* Corresponding author.

E-mail address: m.palatnikov@ksc.ru (M.N. Palatnikov).

<https://doi.org/10.1016/j.ssi.2021.115651>

Received 30 December 2020; Received in revised form 4 March 2021; Accepted 5 May 2021

Available online 9 May 2021

0167-2738/© 2021 Elsevier B.V. All rights reserved.

this case ($\text{Nb}_{\text{Nb}}^{4+} - \text{Nb}_{\text{Li}}^{4+}$) bipolarons make the dominant contribution to conductivity.

I – V curves research made at E close to E_c can provide additional information about the conductivity mechanisms and domain configuration changes in LN crystals. Such a study can also reveal the stability of LN crystals electrical properties under extreme conditions. This topic has been under-estimated, thus we believe that it is important to study and compare I – V curves and conductivity of CLN and LN:Mg and LN:Zn crystals at 300–450 K.

2. Samples and methods

Main experiments were carried out on three LN samples: nominally pure CLN, LN:Mg (4.7 mol% MgO in a crystal), LN:Zn (5.2 mol% ZnO in a crystal). All crystals were grown by Czochralski in ICT RAS (Russian Federation) and were turned to a single-domain state. The samples preparation and the applying measuring electrodes method is described in [16]. The temperature dependence of the specific electrical conductivity $\sigma(T)$ in the temperature range (300–450 K) was studied in all samples. A constant electric field was applied along the polar axis of the crystal Z during the measurements. The temperature dependence of the uncorrected primary pyroelectric coefficient was also measured in order to evaluate the crystals' unipolarity degree; the method is given in [17].

I – V curves research in LN crystals at a fixed temperature has a number of features. Even small changes in temperature lead to pyroelectric currents, which can significantly distort conductivity measurements. Therefore, the measurements were carried out using an inertial type heater with a stabilized supply voltage. The drift rate of the sample temperature V during one measurement did not exceed 0.007 K/min. Real I – V curves researches were carried out at temperatures close to 400 and 450 K with the subsequent normalization to the corresponding temperatures. The normalization procedure will be outlined below.

3. Results and discussion

I – V curves were studied under static electric field of different polarity. This is why correct analysis can be carried out only taking into consideration the initial unipolarity degree of samples. Unipolarity ξ of LN crystals is usually determined as:

$$\xi = \frac{V_+ - V_-}{V_+ + V_-} = \frac{V_+ - V_-}{V_0} \quad (1)$$

where V_+ – crystal volume filled with conditionally positive domains, V_- – conditionally negative domains, V_0 – the total crystal volume [18].

The unipolarity degree can be determined by classical optical methods [2] or as a result of pyroelectric measurements. Obviously, if a completely unipolar crystal has the primary pyroelectric coefficient value (γ_1), at a fixed temperature its value is $(\gamma_1)_m$, then for a multidomain crystal this value will be $(\gamma_1)_p = \xi (\gamma_1)_m$.

Fig. 1 presents a temperature dependence of uncorrected primary pyroelectric coefficient γ' of a sample LN:Zn (5.2 mol% ZnO). This dependence can obviously be approximated by a polynome $\gamma'(T) = \gamma'_{300K} + b(T-300)$. Corresponding measurements (not shown in Fig. 1) were carried out for a nominally pure CLN and LN:Mg (4.7 mol%) samples. Obtained polynomes coefficients are demonstrated in Table 1. The table also shows known from the literature results for other turned to single domain state LN samples [17,19,20]. Table shows that $\gamma'_{300K} +$ values in studied samples coincide within error with known values in other turned to single domain state LN samples [17,19,20]. This indicates a high enough degree of unipolarity of studied crystals. The effect of the samples unipolarity degree on γ' value is clearly demonstrated by the results obtained for the obviously polydomain crystal LN:Zn (2.3 mol% ZnO); these results are also demonstrated on Fig. 1.

The coefficient b means rate of γ' change: $b = \partial\gamma'/\partial T$; for LN:Mg and LN:Zn crystals it differs by almost 2 times. This indicates different

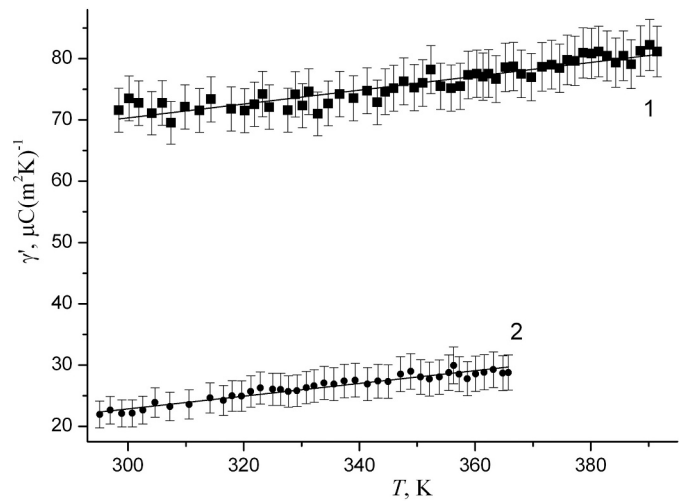


Fig. 1. Temperature dependence of the uncorrected primary pyroelectric coefficient γ' : (1) - in a single-domain LN:Zn sample (5.2 mol%), (2) - in a multi-domain LN:Zn sample (2.3 mol%).

Table 1
Pyroelectric properties of the studied LN crystals.

Sample	$\gamma'_{300K}, \mu\text{C}/(\text{m}^2 \cdot \text{K})^{-1}$	$b, \mu\text{C}/(\text{m}^2 \cdot \text{K}^2)^{-1}$	Temperature range	Ref.
LN:Zn 5.2 mol %	70 ± 4	0.113 ± 0.006	298–390 K	Our data
LN:Mg 4.7 mol%	67 ± 4	0.27 ± 0.01	296–368 K	Our data
CLN	75 ± 4	0.24 ± 0.015	297–380 K	Our data
CLN	65 ± 6	0.34 ± 0.03	300–528 K	[19]
CLN	100 ± 5	0.133	300–420 K	[17]
NSLN	76 ± 4	0.105 ± 0.007	300–420 K	[17]
LN:Zn 6 mol%	70 ± 5	–	–	[20]
LN:Mg 5.0 mol%	66 ± 5	–	–	[20]

temperature dynamics of the cations displacement relative to the oxygen sublattice in these samples.

Fig. 2 demonstrates experimental $\sigma(T)$ dependences of LN:Zn (5.2

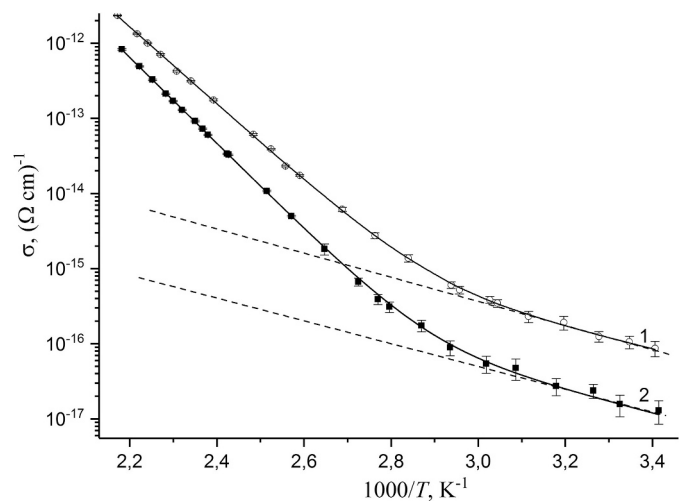


Fig. 2. Temperature dependence of the specific electrical conductivity $\sigma(T)$ along the polar direction z of the samples CLN (1) and LN:Zn (2). The dotted lines indicate the electronic contribution to the conductivity.

mol% ZnO) and CLN crystals. A $\sigma(T)$ dependence of LN:Mg sample is close to that of LN:Zn (not shown in Fig. 2). The temperature dependence of the electrical conductivity $\sigma(T)$ of LN crystals is usually described by Arrhenius law [16]:

$$\sigma(T) = \frac{C_i}{T} \exp\left(-\frac{E_i}{k_0 T}\right) + C_e \exp\left(-\frac{E_e}{k_0 T}\right) \quad (2)$$

where k_0 – Boltzmann constant, T – temperature, E_i and E_e – activation energies of ions and electrons, C_i and C_e – coefficients that depend on the volume concentration and mobility of the corresponding type of carriers.

The parameters corresponding to the optimal approximation of $\sigma(T)$ by expression (2) for the studied samples are given in Table 2. Activation energies of the electron component of conductivity E_e are close to each other and correspond to polaron conductivity activation energy [21]. It is curious that the activation energy E_i for the studied samples is significantly different, Table 2. This may indicate the presence of two types of mobile ions with different combinations of their volume concentration and mobility in different samples [22]. Fig. 2 shows that ionic conductivity makes the main contribution to electrical conductivity of studied samples at a temperature $T_f = 450$ K. Note that activation energy E_i of doped LN:Zn and LN:Mg crystals correspond to Li^+ conductivity [23,24], and of CLN – to proton conductivity [25–27].

At this stage, we have assumed that the activation energies of the conductivity of the studied samples and the C_e coefficient do not depend on the strength of the external electric field E . Thus, it was initially assumed that the external electric field can only affect the mobility of ions – that is, the C_i coefficient.

Normalization of measurement results to a fixed temperature T_f with small deviations (± 2 K) of the real temperature T_s of the sample was carried out with the following procedure. Coefficient C_i' responsible for the ionic contribution to the conductivity was calculated using expression (2) after measuring the conductivity $\sigma(T_s)$ at temperature T_s :

$$C_i' = \frac{\sigma(T_s) - C_e \exp\left(-\frac{E_e}{k_0 T_s}\right)}{\exp\left(-\frac{E_i}{k_0 T_s}\right)} T_s \quad (3)$$

after which the conductivity was reduced to temperature T_f :

$$\sigma(T_f) = \frac{C_i'}{T_f} \exp\left(-\frac{E_i}{k_0 T_f}\right) + C_e \exp\left(-\frac{E_e}{k_0 T_f}\right) \quad (4)$$

After performing the main series of experiments for each sample, a repeated measurement of the conductivity was carried out at low E values. In this case, values of $\sigma(T)$, obtained at the beginning of the measurement cycle, coincided within the error with the values obtained at the end of the cycle. This indicates the long-term stability of the area of electrodes deposited on the crystal and the correctness of experiments.

I – V curves of studied samples at $T = 450$ K are demonstrated on Figs. 3 and 4. At this, the I – V curve becomes nonlinear for a CLN crystal starting from $E_z \cong 10$ kV/mm, Fig. 3. However, at the same temperature and close values of E_z , there is no nonlinearity in the LN:Zn and LN:Mg

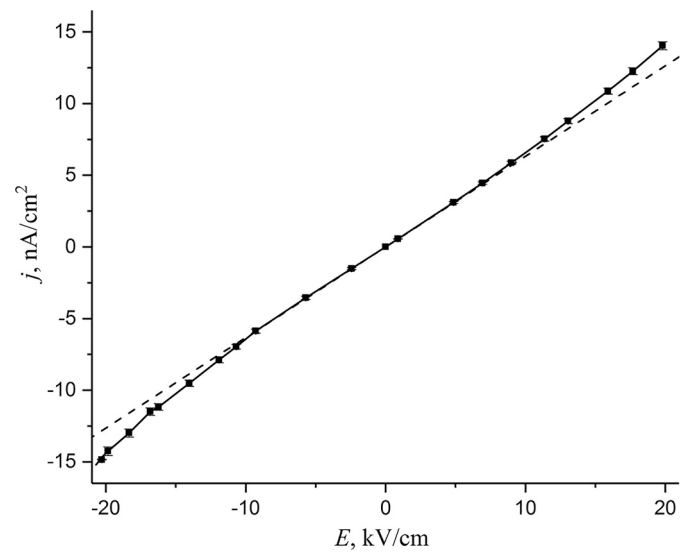


Fig. 3. I – V curve of the CLN sample at $T = 450$ K.

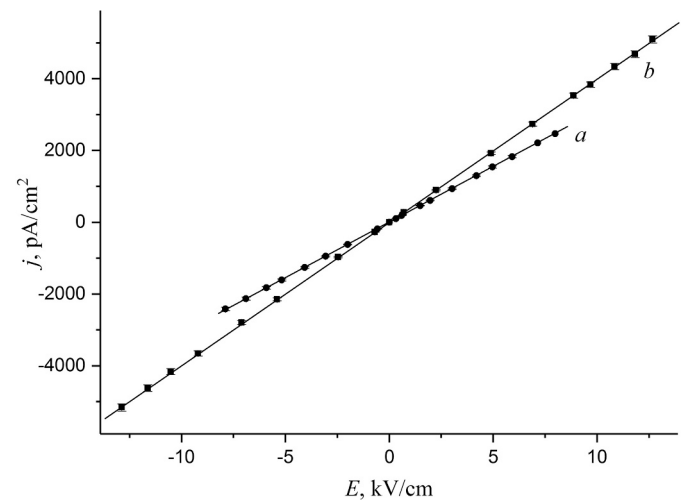


Fig. 4. I – V curve of (a) LN:Mg and (b) LN:Zn crystals at $T = 450$ K.

samples, despite the significantly lower expected E_e value, Fig. 4.

Note that the experiments carried out at $T = 400$ K also failed to show the nonlinearity of the I – V curves of the LN:Zn and LN:Mg samples. This fact indirectly indicates that the electronic contribution to the conductivity is linear in the studied range of values of the external electric field strength.

Possible explanations for the observed I – V curve nonlinearity in CLN crystals and its absence in LN:Mg and LN:Zn crystals are not entirely unambiguous. This phenomenon can manifest itself both in the implementation of one or several probable mechanisms.

First of all, when explaining the phenomenon, it is necessary to understand that switching of spontaneous polarization is not the only and not even the most probable cause of I – V curve nonlinearity. Otherwise, the nonlinearity of the I – V curve would be observed only in crystals in which switching is possible, that is, in ferroelectrics. Thus, apparently, there is no direct relationship between the nonlinearity of the I – V curve and the magnitude of the crystals coercive field E_c . If we nevertheless relate the nonlinearity of the I – V curve with the switching processes, then a bell-shaped pulse of the current density would be observed on the I – V curve with the complete switching of the polarization of the crystal. With partial switching, we could observe the flat wing of this pulse in the form of a deviation from the linear law, similar to what we actually

Table 2

Results of approximation of the dependences $\sigma(T)$ of the investigated LN samples by expression (2).

Sample	C_i , K·(Ω·cm) ^{−1}	E_i , eV	C_e , (Ω·cm) ^{−1}	E_e , eV	σ_{450K} , (Ω·cm) ^{−1}
CLN	$(0.41 \pm 0.08) \cdot 10^{-3}$	1.06 ± 0.01	$(4.94 + 0.08) \cdot 10^{-12}$	0.28 ± 0.03	$(1.24 \pm 0.02) \cdot 10^{-12}$
LN:Mg	$(12.3 \pm 1.8) \cdot 10^{-3}$	1.23 ± 0.01	$(9.5 + 1.2) \cdot 10^{-12}$	0.34 ± 0.04	$(3.07 \pm 0.07) \cdot 10^{-13}$
LN:Zn	$(3.09 \pm 0.33) \cdot 10^{-3}$	1.18 ± 0.01	$(1.73 + 0.17) \cdot 10^{-12}$	0.30 ± 0.02	$(3.98 \pm 0.07) \cdot 10^{-13}$

observe in the CLN crystal, Fig. 3. But, apparently, in the case of $I - V$ curve nonlinearity due to switching processes, nonlinearity would first of all be observed for LN:Zn and LN:Mg crystals, which have significantly lower coercive field E_c values than CLN crystals. In addition, the polarization switching would be accompanied by noticeable Barkhausen effect, which we do not observe in the experiment, Figs. 3 and 4.

On the other hand, it is clear that the linear manifestation of Ohm's law (including the linearity of the $I - V$ curve) is only a special case for relatively weak electric fields. In the general case, we have a power series, the higher terms of which begin to manifest themselves in the experiment at sufficiently high electric field strengths. In this case, the deviation from the linear law may look like the nonlinearity of the $I - V$ curve observed in our experiment, Fig. 3. Moreover, such nonlinearity will begin to manifest itself at lower fields with an increase in crystal conductivity. It was noted in [28] that the conductivity of heavily doped LN:Zn and LN:Mg in the temperature range ~ 310 –500 K is significantly lower than that of CLN crystals.

A decrease in the LN:Zn and LN:Mg crystals conductivity is due to a fundamental rearrangement of the LN crystals structure at high concentrations of magnesium and zinc [2]. At relatively low dopant concentrations, the main acceptor centers (transition metals (Fe, Co...) and Nb_{Li} antisite defects) in LN:Zn and LN:Mg crystals act as electron traps [2]. In this case a re-charge of iron ions $Fe^{3+} + e^- = Fe^{2+}$ can happen or polarons can form upon capture of an electron by an antisite defect Nb_{Li} . These defects participate in the hopping electron or polaron conductivity of LN:Zn and LN:Mg crystals. At high impurity concentrations, as in our LN:Zn and LN:Mg crystals, the Fe^{3+} cations, for example, are localized in the Nb-sites of the structure [29]. It is obvious that the $[Fe^{3+}_{Nb}]^{2-}$ center formed in this process cannot be an electron trap and participate in the electronic conductivity of the crystal. At the same time, such electron traps as Nb_{Li} antisite defects are practically absent in the studied range of impurity concentrations in LN:Zn and LN:Mg crystals [2].

Our study of the LN:Zn and LN:Mg crystals conductivity has fully confirmed the conclusions of [28]. Indeed, the conductivity of CLN crystals in the studied temperature range, including at $T_f = 450$ K, is ~ 3 –4 times higher than that of doped LN:Zn and LN:Mg crystals, Fig. 2 and Table 2. Besides, it was shown above that CLN and LN:Zn, LN:Mg crystals ionic conductivity is different (proton and lithium); accordingly, there is a difference in the mobility and volume concentration of their charge carriers. In addition to noticeably different values of conductivity at $T = 450$ K, these crystals can differ significantly in the values of the coefficients at the higher-order terms of the power-law dependence of the current on the applied electric field. The combination of these facts indicates the possibility of the $I - V$ curve nonlinearity in CLN crystals at significantly lower electric field strengths than in LN:Zn and LN:Mg crystals.

Our paper studies LN crystals turned to a single-domain state with a sufficiently high unipolarity degree, Fig. 1 and Table 1. At the same time, even in LN crystals turned to a single-domain state, there is always a residual, so-called “stubborn” domain structure [30,31]. Moreover, these “stubborn” domains are usually wedge-shaped blind with charged oblique walls [31]. It is known [32] that a number of anomalies of various physical properties are observed in CLN crystals in the practically significant temperature range (300–450 K). For example, anomalous temperature dependences of some optical, dielectric, pyroelectric properties and conductivity are observed, as well as characteristic temperature evolutions of polarization-optical images associated with the rearrangement of the domain structure of crystals. Thus, the reorganization processes of the domain structure can be observed in CLN in the temperature range of ~ 300 –450 K, even without the external electric field [32]. Moreover, in this case we are not talking about the spontaneous polarization switching: the configuration of domains only slightly changes, domain walls insignificantly move, etc. Such reorganizations of the domain structure are clearly visible in polarized light in polished CLN crystal plates placed on the heating stage of a

polarizing microscope. This reorganization will manifest itself in the external electrical circuit as the movement of a charge (in this case, it does not matter whether it is bound or free). Such a reorganization of “stubborn” domains in a certain temperature range can make an additional contribution to the electric current and, accordingly, to the nonlinearity of the CLN crystal $I - V$ curve. Earlier, we have found a similar reorganization of the “residual” domain structure in LN:Zn and LN:Mg crystals, but at much higher temperatures ~ 760 –840 K [28,30,33–37]. Higher temperatures of the effect manifestation in LN:Zn and LN:Mg crystals can be caused by the formation of polar clusters formed by a combination of intrinsic and impurity point defects in strongly doped LN crystals [2]. Such polar clusters can significantly stabilize domain walls [30,33–37] and prevent a change in the domain structure configuration in the range of relatively moderate temperatures of ~ 300 –450 K, Fig. 4. Thus, the residual domain structure in LN:Zn and LN:Mg crystals turned to a single-domain state cannot affect the nonlinearity of the $I - V$ curve, at least in the studied temperature range ~ 300 –450 K, Fig. 4.

It was shown above that doped and CLN crystals at $T = 450$ K have different types of ionic conductivity, lithium and proton, respectively. In addition, the conductivity of CLN crystals at $T = 450$ K is 3–4 times higher than that of LN:Zn and LN:Mg, Fig. 2 and Table 2. Accordingly, the volume concentration and mobility of charge carriers should be different in these crystals. Such differences in the energy and kinetic characteristics of the charge transport process in the region of ionic conductivity (Li^+ and H^+) and when using electrodes irreversible for lithium and hydrogen and measurements at a constant current should lead to different conditions for recharging of conduction cations at the sample cathode. First of all, it should be noted that at the studied values of cationic conductivity no polarization of the electrodes irreversible with respect to lithium and hydrogen occurs. Otherwise, we would observe a decrease in the electric current through the sample at long measurement times, which is actually not observed in the experiment, Fig. 3. Moreover, with an increase in the electric field strength, the current through the CLN crystal sample increases, Fig. 3. In this case, when the conduction cation approaches the cathode to neutralize its charge, even at a relatively low temperature $T = 450$ K, a process that looks similar to thermionic emission from a metal electrode should take place. Thermionic emission at sufficiently high values of ionic conductivity can contribute to the measured current and, accordingly, the nonlinearity of the $I - V$ curve. It should be noted that any metal electrodes are quite easily permeable for protons, and such a charge exchange in CLN crystals can occur more efficiently than in LN:Zn and LN:Mg crystals and, therefore, can contribute to the nonlinearity of the $I - V$ curve. In addition, when measured in air, Li^+ cations, upon reaching the cathode, can interact with oxygen to form lithium oxide. In this case, the chemical neutralization of the cation charge will occur without the contribution of this process to the through current and, accordingly, to the nonlinearity of the $I - V$ curve.

Thus, there is no unified and comprehensive explanation for the nonlinearity of the $I - V$ curve in the CLN crystal. The nonlinearity can be due to the action of one or several of the proposed mechanisms at the same time. Moreover, the least probable of them is the spontaneous polarization switching of the crystalline sample.

4. Conclusion

The conductivity and current-voltage characteristics were investigated in pure CLN and LN:Zn, LN:Mg crystals turned to a single domain state at temperatures $T = 300$ –450 K. The magnitude, activation energy, and type of electronic and ionic conductivity in CLN and LN:Zn, LN:Mg crystals were determined.

In CLN crystals, the coercive field E_c is significantly higher than in LN:Zn and LN:Mg crystals. Despite this, the CLN crystal exhibits the effect of nonlinearity of the $I - V$ curve at a temperature of 450 K and fields up to 10 kV/mm. At this, the nonlinearity is absent in LN:Zn and

LN:Mg crystals at external electric field strength up to 12 kV/mm and $T = 450$ K.

Apparently, the effect of the $I - V$ curve nonlinearity in the CLN crystal is not associated with the spontaneous polarization switching, otherwise it would first of all be observed in LN:Zn and LN:Mg crystals, which have a lower coercive field. In addition, the experiment was not accompanied by Barkhausen effect, which is usually observed at spontaneous polarization switching. Possible mechanisms of manifestation of the observed effects are considered in this work.

Data availability

The data will be available from the corresponding author upon a reasonable response.

Author statement

We, A. V. Yatsenko, S.V. Evdokimov, M. N. Palatnikov, N. V. Sidorov, authors of paper “Analysis of the conductivity and current-voltage characteristics nonlinearity in LiNbO₃ crystals of various compositions at temperatures 300 - 450 K”, confirm that

- the work described has not been published previously;
- it is not under consideration for publication elsewhere;
- its publication is approved by all authors and tacitly or explicitly by the responsible authorities where the work was carried out;
- we agree to publish this paper in the journal “Solid state ionics”;
- if accepted, it will not be published elsewhere in the same form, in English or in any other language, including electronically without the written consent of the copyright-holder.

Funding

This work was supported by the Ministry of Science and Higher Education Russian Federation scientific topic No 0186-2021-0022 (registration # AAAA-A21-118022190125-2).

Declaration of Competing Interest

The authors declare no conflict of interest.

References

- [1] L. Arizmendi, Photonic applications of lithium niobate crystals, *Phys. Stat. Sol. (a)*. 201 (2) (2004) 253–283, <https://doi.org/10.1002/pssa.200303911>.
- [2] T. Volk, M. Wöhlecke, Lithium Niobate. Defects, Photorefractive and Ferroelectric Switching, Springer, Berlin, 2008, p. 250, <https://doi.org/10.1007/978-3-540-70766-0>.
- [3] Y. Chen, W. Yan, J. Guo, S. Chen, G. Zhang, Effect of Mg concentration on the domain reversal of Mg-doped LiNbO₃, *Appl. Phys. Lett.* 87 (2005) 212904, <https://doi.org/10.1063/1.2135389>.
- [4] L.-H. Peng, Y.-C. Zhang, Y.-C. Lin, Zinc oxide doping effects in polarization switching of lithium niobate, *Appl. Phys. Lett.* 78 (4) (2001) 4, <https://doi.org/10.1063/1.1336815>.
- [5] C.T. Chia, C.C. Lee, P.J. Chang, M.L. Hu, L.J. Hu, Substitution mechanism of ZnO-doped lithium niobate crystal determined by powder x-ray diffraction and coercive field, *Appl. Phys. Lett.* 86 (2005) 182901, <https://doi.org/10.1063/1.1922083>.
- [6] H. Ishizuki, I. Shoji, T. Taira, Periodical poling characteristics of congruent MgO:LiNbO₃ crystals at elevated temperature, *Appl. Phys. Lett.* 82 (2003) 4062, <https://doi.org/10.1063/1.1582371>.
- [7] H. Steigerwald, F. von Cube, F. Luedtke, V. Dierolf, K. Buse, Influence of heat and UV light on the coercive field of lithium niobate crystals, *Appl. Phys. B Lasers Opt.* 101 (2010) 535–539, <https://doi.org/10.1007/s00340-010-4209-1>.
- [8] V.Ya. Shur, A.R. Akhmatkhanov, I.S. Baturin, Micro- and nano-domain engineering in lithium niobate, *Appl. Phys. Rev.* 2 (2015) 040604, <https://doi.org/10.1063/1.4928591>.
- [9] B. Chikh-Bled, B. Benyoucef, M. Aillerie, Experimental measurement of electric conductivity and activation energy in congruent lithium niobate crystal, *J. Active Passive Electron. Dev.* 7 (3) (2012) 261–270.
- [10] B. Chikh-Bled, M. Aillerie, B. Bouncef, Dark and photo-conductivity measurement techniques for dielectric materials, application to LiNbO₃, *J. Eng. Appl. Sci.* 6 (2) (2011) 163–167, <https://doi.org/10.3923/jeasci.2011.163.167>.
- [11] K.S. Tunik, A.V. Yatsenko, Investigation of volt-ampere characteristics of nominally pure congruently grown LiNbO₃ crystals, *J. Phys. Technol. Inst. V.I. Vernadsky Crimean Federal Univ.* 1 (67–69) (2017) 50–55 (in Russian).
- [12] N. Meyer, G.F. Nataf, T. Granzow, Field induced modification of defect complexes in magnesium-doped lithium niobate, *J. Appl. Phys.* 116 (2014) 244102, <https://doi.org/10.1063/1.4905021>.
- [13] L. Jentjens, N.L. Raeth, K. Peithmann, K. Maier, Electrical conductivity and asymmetric material changes upon irradiation of mg-doped lithium niobate crystals with low-mass, high-energy ions, *J. Appl. Phys.* 109 (2011) 124104, <https://doi.org/10.1063/1.3596824>.
- [14] P.F. Bordui, D.H. Jundt, E.M. Standifer, R.G. Norwood, Chemically reduced lithium niobate single crystals: processing, properties and improved surface acoustic wave device fabrication and performance, *J. Appl. Phys.* 85 (7) (1999) 3766, <https://doi.org/10.1063/1.369775>.
- [15] L.D. Kaverin, L.N. Korotkov, Conductivity of single crystal LiNbO₃ annealed in H₂ atmosphere, *Ferroelectrics*. 214 (1998) 333–337, <https://doi.org/10.1080/00150199808220274>.
- [16] A.V. Yatsenko, M.N. Palatnikov, N.V. Sidorov, A.S. Pritulenko, S.V. Evdokimov, Specific features of electrical conductivity of LiTaO₃ and LiNbO₃ crystals in a temperature range of 290–450 K, *Phys. Solid State* 57 (8) (2015) 1547–1550, <https://doi.org/10.1134/S1063783415050339>.
- [17] A.V. Yatsenko, S.V. Evdokimov, M.N. Palatnikov, N.V. Sidorov, Features of the electrical properties of lithium niobate crystals grown from a melt containing K₂O flux, *Phys. Solid State* 61 (7) (2019) 1211–1217, <https://doi.org/10.1134/S106378341907031X>.
- [18] M.N. Palatnikov, V.A. Sandler, N.V. Sidorov, I.V. Biryukova, O.B. Makarova, Physicochemical, dielectric, and piezoelectric properties and conductivity of LiNbO₃:ZnO crystals (4.02–8.91) mol (%), *Techn. Phys.* 62 (1) (2017) 82–89, <https://doi.org/10.1134/S1063784217010170>.
- [19] T. Bartolomäus, K. Buse, C. Deuper, E. Krätzig, Pyroelectric coefficients of LiNbO₃ of different compositions, *Phys. Stat. Sol. A*. 142 (1) (1994) K55–K57, <https://doi.org/10.1002/pssa.2211420146>.
- [20] S.L. Bravina, N.V. Morozovskaya, A.N. Morozovskaya, S. Gille, J.-P. Salvestrini, M. D. Fontana, Investigation of LiNbO₃ and LiTaO₃ single crystals for pyroelectric applications in the wide temperature range, *Ferroelectrics* 353 (1) (2007) 202–211, <https://doi.org/10.1080/00150190701368166>.
- [21] A.V. Yatsenko, S.V. Evdokimov, Effect of an iron impurity on the electrical conductivity of LiNbO₃ crystals, *Phys. Solid State* 62 (2020) 485–491, <https://doi.org/10.1134/S1063783420030269>.
- [22] A.S. Pritulenko, A.V. Yatsenko, S.V. Evdokimov, Analysis of the nature of electrical conductivity in nominally undoped LiNbO₃ crystals, *Crystallogr. Reports*. 60 (2) (2015) 267–272, <https://doi.org/10.1134/S1063774515020224>.
- [23] G.T. Niitsu, H. Nagata, A.C.M. Rodríguez, Electrical properties along x and z axes of LiNbO₃ wafers, *J. Appl. Phys.* 95 (2004) 3116–3119, <https://doi.org/10.1063/1.1647263>.
- [24] B. Ruprecht, J. Rahn, H. Schmidt, P. Heitjans, Low-temperature DC conductivity of LiNbO₃ single crystals, *Z. Phys. Chem.* 226 (2012) 431–437, <https://doi.org/10.1524/zpch.2012.0226>.
- [25] E.M. de Miguel-Sans, M. Carrascosa, L. Arizmendi, Effect of oxidation state and hydrogen concentration on the lifetime of thermally fixed holograms in LiNbO₃:Fe, *Phys. Rev. B* 65 (2002) 165101, <https://doi.org/10.1103/PhysRevB.65.165101>.
- [26] B. Chikh-Bled, B. Benyoucef, M. Aillerie, Experimental measurement of electric conductivity and activation energy in congruent lithium niobate crystal, *J. Active Passive Electron. Dev.* 7 (2012) 261–270, <https://doi.org/10.1063/1.3596824>.
- [27] A.V. Yatsenko, A.S. Pritulenko, S.V. Evdokimov, D.Yu. Sugak, I.M. Solskii, The peculiarities of the electrical conductivity of LiNbO₃ crystal reduced in hydrogen, *Sol. St. Phen.* 200 (2013) 193–198, <https://doi.org/10.4028/www.scientific.net/SSP.200.193>.
- [28] M.N. Palatnikov, V.Ya. Yatsenko, V.A. Sandler, N.V. Sidorov, O.V. Makarova, D. V. Manukovskaya, Dielectric properties and electrical conductivity of LiNbO₃:Zn crystals in the temperature range 310–900 K, *Sol. State Ion.* 345 (2020) 115178, <https://doi.org/10.1016/j.ssi.2019.115178>.
- [29] H. Feng, J. Wen, H. Wang, Sh. Han, Yu. Xu, EPR studies in Mg-doped LiNbO₃ crystals, *J. Phys. Chem. Solids* 51 (1990) 397–400, [https://doi.org/10.1016/0022-3697\(90\)90173-D](https://doi.org/10.1016/0022-3697(90)90173-D).
- [30] M.N. Palatnikov, V.A. Sandler, O.V. Makarova, N.V. Sidorov, D.V. Manukovskaya, I.N. Efremov, I.V. Biryukova, K. Bormanis, Anomalies of dielectric, piezoelectric properties and conductivity in highly doped single domain LiNbO₃:Zn crystals, *Integr. Ferroelectr.* 173 (2016) 119–127, <https://doi.org/10.1080/10584587.2016.1186431>.
- [31] V.Ya. Shur, E.L. Rumyantsev, E.V. Nikolaeva, E.I. Shishkin, Formation and evolution of domain walls in congruent lithium niobate, *Appl. Phys. Lett.* 77 (2000) 3636–3638, <https://doi.org/10.1063/1.1329327>.
- [32] A.M. Prokhorov, Yu.S. Kuz'minov, *Physics and Chemistry of Crystalline Lithium Niobate*, Adam Hilger, New York, 1990, p. 237.
- [33] M.N. Palatnikov, V.A. Sandler, N.V. Sidorov, O.V. Makarova, Anomalous dielectric and piezoelectric properties and electrical conductivity of heavily doped LiNbO₃:Zn crystals, *Inorg. Mater.* 52 (2016) 147–152, <https://doi.org/10.1134/S0020168516020114>.
- [34] M.N. Palatnikov, V.A. Sandler, N.V. Sidorov, I.V. Biryukova, O.V. Makarova, Dielectric and piezoelectric properties and electrical conductivity of LiNbO₃:Zn crystals in wide range of dopant concentration, *Inorg. Mater.* 52 (12) (2016) 1291–1296, <https://doi.org/10.1134/S0020168516120086>.
- [35] M.N. Palatnikov, V.A. Sandler, N.V. Sidorov, I.V. Biryukova, O.V. Makarova, Physicochemical, dielectric, and piezoelectric properties and conductivity of

- LiNbO₃:ZnO crystals (4.02–8.91 mol%), Techn. Phys. 62 (2017) 82–89, <https://doi.org/10.1134/S1063784217010170>.
- [36] M.N. Palatnikov, V.A. Sandler, N.V. Sidorov, O.V. Makarova, Evolution of the domain structure of LiNbO₃:ZnO crystals during high-temperature annealing, Inorg. Mater. 54 (9) (2018) 915–919. <https://doi.org/10.1134/S002016851809011X>.
- [37] M.N. Palatnikov, V.A. Sandler, N.V. Sidorov, O.V. Makarova, Mechanisms of variation of the unipolarity during thermal processing of heavily doped LiNbO₃:ZnO crystals, Techn. Phys. 65 (2020) 1246–1252, <https://doi.org/10.1134/S1063784220080149>.